

18 Answer: B

$$v_X = u_X = 3.1 \times 10^7 \text{ m s}^{-1}$$

$$s_X = u_X t$$

$$t = 0.12 / 3.1 \times 10^7 = 3.8710 \times 10^{-9}$$

$$a_Y = F / m = 3.2 \times 10^{-15} / (9.11 \times 10^{-31}) = 3.5126 \times 10^{15}$$

$$v_Y = u_Y + a_Y t = 0 + (3.5126 \times 10^{15}) (3.8710 \times 10^{-9})$$

$$v_Y = 1.3597 \times 10^7 \text{ m s}^{-1}$$

$$\tan \theta = v_X / v_Y$$

$$\theta = 24^\circ$$